

Biotechnology and Society BSCI 422 Fall 2017

Professor	Dr. Sue Cooke
Email	Suzanne.Cooke@unh.edu
Office Hours	Wednesdays 12-1 (on Manchester campus) or by appointment
Class Dates	October 23 rd - December 15 th
Textbook	<i>none</i>

Format: Online Asynchronous. Each week, a new module including lecture material and assignments will be posted.

Readings:

There is no required textbook for this course. The reading assignments and online articles will be listed in within each module on MyCourses.

Expectations:

This class moves at twice the pace of a regular full term course. Stay on top of the work and don't fall behind. "Attendance" is mandatory in that you are expected to log in weekly to complete the assigned work. Due to the accelerated nature of the course and the online format, due dates are firm and late work will be penalized with a loss of 10% per day. All online discussions have three requirements (see below) to receive full credit. I will be corresponding with you throughout the term via UNH email (the email address listed for you in MyCourses). Check your UNH email daily for updates and communication about this course.

Grading:

You will be graded out of a total of 350 points, broken up as follows:

Weekly Discussion Board Assignments (20pts each x 6)	120
News Article Assignment:	
Weekly comments (10pts each x 5)	50
Provide article and lead discussion (1x during course)	30
Research Abstract Assignment:	
Library homework	10
Topic submission	10
Annotated Bibliography	30
CAE review of abstract	10
Research Abstract	50
Video Presentation of Abstract	30
Feedback on peers' presentations	10
Total	350

Assignments:

Weekly Topic Discussion Board assignments:

To get full credit for the discussion you need to complete the following by the deadlines.

1. Post your initial comment on the topic not later than **midnight on Friday** of the assigned week. In your comment, please provide a link to an online article that is relevant to the topic being discussed and pose a question to the class related to your article. (5pts)
2. Respond to at least two of your peers' posts by **midnight on Sunday** of the assigned week. (10pts)
3. Respond to any comments you receive on your own post by **midnight on Sunday** of the assigned week. (5pts)

When selecting articles to share with the class, please keep in mind there is a lot of junk out there on the internet and anyone can post an article! Stick with newspapers, periodicals, journals and reputable scientific sites. By posting a link to an article you draw the class into the discussion of what you found by posing a question. Do not just post a link/URL without a question. The idea is to give current information on the subject, create more interest in the topic and generate a robust discussion.

News Article Assignment

You are required to share one current news article from a reputable source pertaining to course material with the class and lead the discussion on that article. You are encouraged to both pose and respond to questions throughout the week. It does not have to be related to the material we are covering that week in class but it does have to be related to Biotechnology in general. I will post a schedule of which students have been assigned to each week. On your assigned week, you will share your article through a discussion board and answer any questions received from other members of the class related to your article. Discussion Board requirements are the same as for the Weekly Discussion Board Assignments.

Research Abstract and Presentation

This is a scaffolded assignment (due dates in course schedule) that requires you to submit your topic, submit an annotated list of references, write an abstract and prepare/record a Powerpoint presentation.

Library videos and homework: A librarian from our library will be providing two videos for you to review. During week 1, you will review a video on the resource guide for this class and suggestions for how to pick a topic for your research abstract. During week 2, you will review a video on finding resources for your abstract. After viewing both videos, the library has provided a brief review assignment that you will need to complete and submit.

Topic: There are so many topics we could have covered in this class it was a difficult decision to choose just a few. Pick a biotechnology topic that interests you that we will not cover in class. Submit your topic via the link provided for me to approve. First come, first serve... no two people can do the same topic! At the end of the course, we will have a mini "research conference". Students will prepare a video presentation to share their topic (virtually) with the class. Presentations should be 8-10 minutes in length (approximately 10-12 slides) and follow the outline of your abstract. You will also be required to view and provide feedback on your peers' presentations.

Annotated bibliography: Submit at least 5 references with authors from journals, science magazines and newspapers. They must be in APA format. This list can be changed after submission if

needed. No textbooks, encyclopedias, dictionaries, popular magazines, thesaurus, podcasts, tweets, videos, blogs or websites will be accepted. Only exception is one reference from Access Science or Britannica if needed for definition purposes. The Access Science and Britannica reference should only be used once in your abstract. For each reference, provide a brief summary in your own words about what it will contribute to your abstract. After you submit your references, I will make corrections if necessary. Be sure to make the corrections I suggest before including them with your abstract.

Abstract: The abstract is one solid single spaced paragraph containing 250-300 words including citations. Do not indent the paragraph. Your heading is Abstract. Use the following outline to write your abstract.

- A. Introduction of topic
- B. Potential uses of technology (include what groups of people will benefit)
- C. Controversies surrounding the use of the technology
- D. Significance of the technology and its future

You may add to or change the references from what was submitted with your annotated bibliography but there must be at least 5 references with authors from journals, science magazines or newspapers. Encyclopedias, thesaurus, books, dictionaries or textbooks cannot be used as references. Exception is one citation from Access Science or Britannica (you should only cite this once in your abstract). Online websites are not acceptable references. Please use APA format for your in-text citations as well.

Use the Center for Academic Enrichment for help writing your abstract and using APA format. Bring them a draft of your abstract by **December 3rd** for review and earn **10 points**. A form will be provided that you will submit to show you completed this requirement. Make your appointment with the science writing tutor today.

Academic Honesty:

The Biology and Biotechnology Programs at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Center for Academic Enrichment (CAE)

The **Center for Academic Enrichment** is a resource for all students enrolled in **UNH Manchester** courses. Our mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. **All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. Make an appointment <https://unh.mywconline.net/> or call 641-4113**

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek who can be contacted at (603) 641-4170, jenessa.zurek@unh.edu or in person in the Student Services Suite, Room (#410H).

Schedule of Topics and Assignments

<i>Week</i>	<i>Date</i>	<i>Topics Covered</i>	<i>Assignments and Due Dates</i>
One	10/23/17 – 10/29/17	Introduction to Biotechnology, Implications of Human Genome Project	• Complete Implications of HGP Discussion Board • Join our class Facebook Page and introduce yourself to the class • Watch 1st library video
Two	10/30/17 – 11/5/17	GMO's & Aquatic Biotechnology, DNA Paternity Testing	• Complete DNA Paternity Testing Discussion Board • Review one student-selected article on our Facebook page and comment/discuss • Watch 2nd library video, complete and submit library homework
Three	11/6/17 – 11/12/17	Stem Cells & Therapeutic Cloning, Assisted Reproduction	• Complete Assisted Reproduction Discussion Board • Review student-assigned news articles and respond • Submit topic for Research Abstract & Presentation
Four	11/13/17 – 11/19/17	Microbial Fermentation & the Human Microbiome, DNA Fingerprinting	• Complete DNA Fingerprinting Discussion Board • Review student-assigned news articles and respond • Submit your Annotated Bibliography for your Research Abstract & Presentation
Five	11/20/17 – 11/26/17	Thanksgiving Week	• No new material this week • Work on your Research Abstract & Presentation
Six	11/27/17 – 12/3/17	Vaccines, 3-D Biotechnologies	• Complete 3-D Biotechnologies Discussion Board • Review student-assigned news articles and respond • Submit documentation of CAE review of abstract
Seven	12/3/17 – 12/10/17	Bioremediation	• Review student-assigned news articles and respond • Submit your Research Abstract & Presentation • Complete Wrap Up Discussion Board
Eight	12/11/17 – 12/15/17	Research Abstract Presentations	• Review your peers' presentations and post feedback on the Feedback Discussion Board

Class Detailed Topic Schedule

Week 1 (Oct. 23-29):

Introduction to Biotechnology

Biotechnology, proteomics and bioinformatics will be defined and applications introduced.

Implications of Human Genome Project

We will discuss how the HGP was completed and the medical, social and ethical impact it will have in our lives.

Week 2 (Oct. 30-Nov. 5):

Genetically Modified Organisms and Aquatic Biotechnology

Proponents of GMO crops say that it takes a reduced amount of pesticide and the yield is greater. Opponents of GMO food say that there could be risk of allergic reactions and the GMO crops could hybridize with wild plants making superweeds. Fish farming increases production of popular fish and lowers the price. Making a transgenic fish can enhance certain qualities yet is there assurance this technology can be contained? We will discuss techniques, benefits and controversies surrounding this technology. We will discuss the benefits and controversies surrounding this technology.

DNA paternity testing

After reading the case of Cheryl from the Massachusetts Supreme Court we will discuss the statute of limitations on questioning paternity. With DNA testing, paternity can now be proven unambiguously. Each state varies in its laws and some have no statute of limitations. If a parent waited five years to question paternity should he still be obligated to pay child support if he is not the father? Is there more to being a parent than DNA? We will discuss where parental obligations begin and end.

Week 3 (Nov. 6-12):

Stem cells and Therapeutic Cloning

Most people don't know the difference between cloning and therapeutic cloning. We will discuss the risks and benefits of each.

Assisted Reproduction

Reproductive technology is a rapidly developing field. Women are having babies in their 60s. Is there equal access to this technology? Who decides if it is a given right to conceive and have a child? Who should pay for these expensive therapies? We will discuss all these issues and the future of these technologies.

Week 4 (Nov. 13-19):

Microbial Fermentation and the Human Microbiome

We have used fermented products for centuries. Think bread, beer and sauerkraut. We will also investigate the new genome project, the recently funded Human Microbiome Project. Other uses for bacteria include production of electricity.

DNA fingerprinting

Your DNA fingerprint can be obtained by taking a sample of your cheek cells and analyzing it in a lab. Once someone has your DNA fingerprint, who else will have access to it? In NH, are there state or federal laws that protect your genetic privacy? We will look at the Innocence project where many

death row inmates have been proven innocent and freed as a result of DNA fingerprint technology. Also how DNA finger printing is used in paternity cases.

Week 5 (Nov. 20-26): Thanksgiving week.

No new material. You will work on your Research Abstract and Presentation.

Week 6 (Nov. 27-Dec. 3):

Vaccines - the Lifesaving Technology

This lifesaving technology has been around for over one century. Yet widespread misinformation has led to a decline in vaccination rates of children. This irresponsibility has led to measles outbreaks and death. Also of interest, it is people who are already fully vaccinated making the decisions for minors that could also detrimentally affect others.

3-D Biotechnologies

While imaging has been common since Röntgen invent the X-ray machine imaging has advanced rapidly over the past thirty years. With advanced computer software and 3-D printers there seems to be no limit to these technologies. Many high schools have 3-D printers. We will discuss the current uses of these technologies and the hopes for the future.

Week 7 (Dec. 4-10):

Bioremediation

Little microbes can do big things too. We will go to Dirtland and learn about the many different ways microbes can clean up toxic waste. We will also look at current research being done that demonstrates that microbes can generate electricity. The future for these technologies looks bright. We will discuss how commonplace they may become.

Week 8 (Dec. 11-15) Research Abstract Presentations

We will host our virtual “mini-conference” in the class this week and you will have the opportunity to view and comment on your peers’ research abstract presentations.

This syllabus is tentative and subject to change.